

ECE 551

HW5

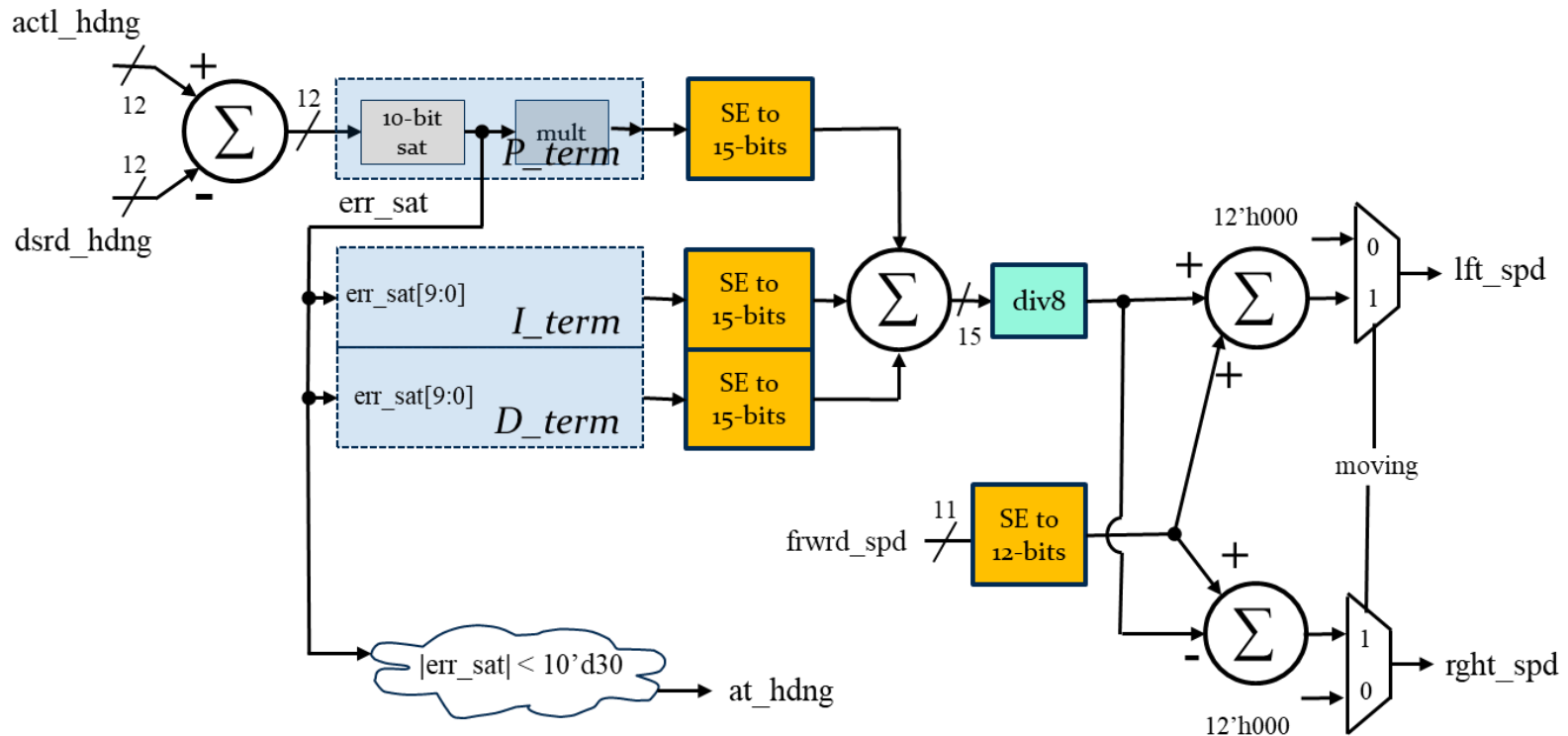
- Due Mon April 20th by 11:55PM
- Work Individually on Problems 1 & 2
- Rest of HW can be done as a team

HW5 Problem 1 (This problem is **individual** basis)

1. (15pts) PID random testing.

Individual problem and not covered by exercise.

Way back in Ex06, Ex08, and Ex10 you created the P,I, and D terms of our PID block, and assembled into PID unit and ran through a provided testbench. That testbench was decent...but not totally thorough. In this exercise you will run your design against provided random vectors and expected response.



HW5 Problem 1 (PID creation & random testing)

- You will test your **PID.sv** unit using stimulus and expected response read from a file. On the website you will find **PID_stim.hex** and **PID_resp.hex**. These represent stimulus and expected response.
- For **PID_stim.hex** the vector is 38 bits wide and is assigned as follows:
- For **PID_resp.hex** the vector is 25 bits wide and is assigned as follows:

Stimulus Bit Range:	Signal Assignment:
stim[37]	rst_n
stim[36]	moving
stim[35]	hdng_vld
stim[34:23]	dsrd_hdng
stim[22:11]	actl_hdng
stim[10:0]	frwrd_spd[10:0]

Resp Bit Range:	Signal Assignment:
resp[24]	at_hdng
resp[23:12]	lft_spd
resp[11:0]	rght_spd

Submit:

- PID.sv
- PID_tb.sv
- Image of transcript window with name visible

- Create **PID_tb.sv** that instantiates your PID and declares a stimulus and resp memory to hold the 2000 vectors of stim & resp. Read each provided file into its respective memory using **\$readmemh**. (**see next 2 slides**)
- Loop through the 2000 vectors and apply the stimulus vectors (*starting with clk low*) to the inputs as specified. Then wait till #1 time unit after the rise of **clk** and compare the DUT outputs to the response vector (self check). Do all 2000 vectors match? If so print a happy message to the transcript window **that includes your name**.

HW5 Problem 1 (Debug...some intermediate values)

I realize debug of random vectors can be difficult. Shown below are some of my intermediate values for the first 12 vectors. This may aid in your debug efforts.

chk_tb/DUT/dk	0																			
chk_tb/DUT/error	3d7	1a9	6b0	3d8	423	8e8	875	2bf	74f	bda	841	6b0	34f	3d7						
chk_tb/DUT/P_term	05fd	04fb	05fd			3a00		05fd		3a00		05fd								
chk_tb/DUT/I_term	05f	000				01f			03f	05f	03f	01f	03f	05f						
chk_tb/DUT/D_term	0000	06f2				1900		06f2	0000	1900		06f2		0000						
chk_tb/DUT/PID	065c	0bed	0cef			731f		0d0e	063c	735f	733f	0d0e	0d2e	065c						
chk_tb/DUT/integrator	05fb	0000				01ff			03fe	05fd	03fd	01fd	03fc	05fb						

Loading Memory Data From Files

Inserted from lecture slides for ease of reference

- This is very useful (memory modeling & testbenches)
 - \$readmemb("<file_name>",<memory>);
 - \$readmemb("<file_name>",<memory>,<start_addr>,<finish_addr>);
 - \$readmemh("<file_name>",<memory>);
 - \$readmemh("<file_name>",<memory>,<start_addr>,<finish_addr>);
- **\$readmemh** → Hex data...**\$readmemb** → binary data
 - But they are reading ASCII files either way (just how numbers are represented)

```
// addr  data
@0000 10100010
@0001 10111001
@0002 00100011
```

example "binary"
file

```
// addr  data
@0000  A2
@0001  B9
@0002  23
```

example "hex"
file

```
//data
A2
B9
23
```

address is optional
for the lazy

Example of \$readmemh

Inserted from lecture slides for ease of reference

```
module rom(input clk; input [7:0] addr; output [15:0] dout);

reg [15:0] mem[0:255];    // 16-bit wide 256 entry ROM
reg [15:0] dout;

initial
    $readmemh("constants",mem);

always @(negedge clk) begin
    //////////////////////////////////////
    // ROM presents data on clock low //
    //////////////////////////////////////
    dout <= mem[addr];
end

endmodule
```

HW5 Problem 2 (This problem is **individual** basis)

2. (20pts) Post synthesis simulation

Individual problem and not covered by exercise.

Posted on the class webpage (under the tutorials section) is a file about Post Synthesis Validation (simulation). Read it carefully.

As part of HW4 (and Ex16) you created a synthesis script to synthesize your **UART**. Adapt that synthesis script to apply to your **PID.sv** block (*but relax clock period to 10ns*). Synthesize and produce **PID.vg**.

Now incorporate that gate level netlist into your test bench (**PID_tb.sv**) and prove your post synthesis netlist works.

Submit:

- a) **PID.vg**
- b) Simulation results proving success.
 - i. Image of your transcript loading the library cells
 - ii. Image of your transcript window showing success

You are submitting images to prove:

- 1.) This is indeed a post synth simulation
- 2.) You indeed ran it (***your name***/username somewhere on image)
- 3.) Evidence it passed (*or not...some credit for trying*)

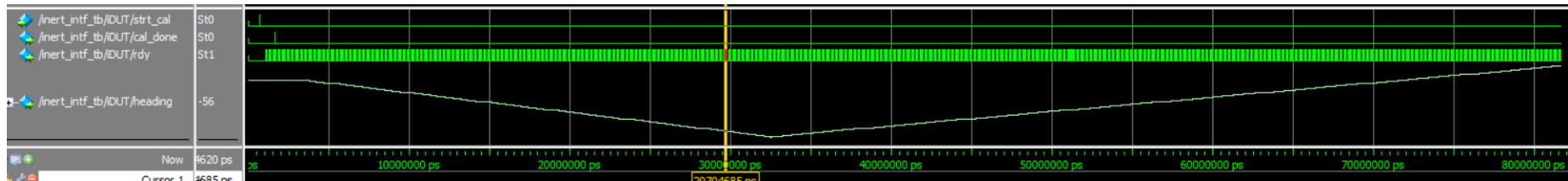
(**Note:** you may want to try post synthesis simulation of a few of your other blocks to ensure they are ready for the project)

HW5 Problem 3 (This problem is done on **Team** basis)

Started as Ex17 on March 23rd

3. (20pts) inertial_interface

From Exercise17 your team should ideally have two implementations of **inert_intf.sv**. Pick one of the implementations for submission for this HW and for use on the project.



Submit: **inert_intf.sv**, **inert_intf_tb.sv**

One person from the group submits & make sure to add everyone into the submission.

Did your team check off for Ex18 (mapping of inert_intf to DE0)? If not, you can get it checked off during office hours still. This is also **required** for full points.

HW5 Problem 4 (This problem is done on **Team** basis)

Started as Ex19 on April 6th

4. (25pts) cmd_proc

See Exercise 19 for specification

Submit:

- cmd_proc.sv
- cmd_proc_tb.sv
- Proof it ran and passed your testbench (make sure your user name can be seen in transcript)

HW5 Problem 5 (This problem is done on **Team** basis)

Started as Ex19 on April 6th

5. (20pts) piezo_drv

See Exercise 19 for specification

This was the 2nd part of Ex19. Did your team check off on piezo_drv on the DE0 nano? If so, you are done with this problem.

Submit:

- piezo_drv.sv
- piezo_drv_tb.sv
- Proof it ran and passed your testbench (make sure your user name can be seen in transcript)
- piezo_drv_test.sv

Partial credits given if your team did not get checked off on the DE0. You can get it checked off during office hours until HW due date.